Corn and Wheat Pairs Trading

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Abstract

This project studies whether the relative price relationship between Corn and Wheat futures contains a small but usable forecasting signal. We built a reproducible pipeline using Bloomberg continuous futures data, constructed a paper-derived mean-reversion feature, and compared Elastic Net with LightGBM using chronological cross-validation. The models were then translated into a hedge-ratio-neutral pairs portfolio and evaluated after transaction costs. The strongest reported specification was an Elastic Net model with a no-trade threshold. On the final holdout period, it produced a net Sharpe ratio of approximately 0.42 at 2 basis points per unit of turnover, with a maximum drawdown of about 7.5 percent. The result is encouraging but not conclusive. Performance varied substantially by year, the active hit rate was slightly above 50 percent, and the deflated Sharpe ratio was approximately zero after accounting for multiple trials.

1 Research Question and Data

We asked a fairly narrow question: can a repeatable signal in the Corn and Wheat price relationship survive the steps between a statistical prediction and an implementable portfolio? Corn and Wheat have a clear economic connection through agricultural production, storage, transportation, and substitution, but their relative price can still move away from its recent historical range.

The primary research dataset is a Bloomberg workbook containing daily continuous futures prices from January 2000 to July 2026. The raw Corn and Wheat series contained 6,836 rows. After cleaning and aligning both legs, the final panel contained 6,692 observations. Yahoo Finance data was also collected and frozen as a provisional comparison. It was useful for the initial pair screen, but it was not used for the final feature engineering or backtest because its continuous-contract construction was less transparent and its history was shorter.

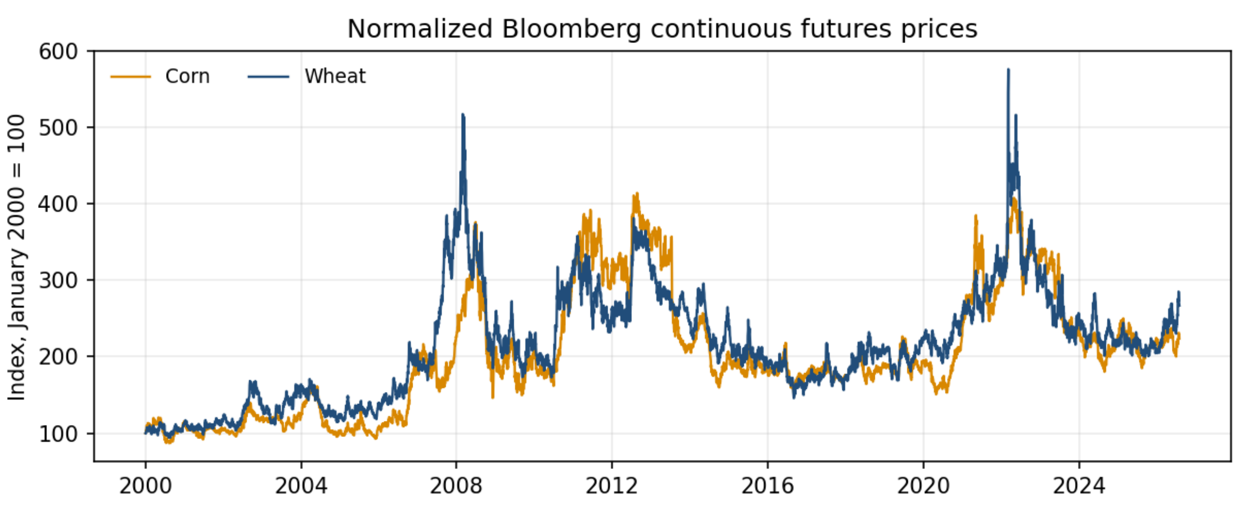

Figure 1: Corn and Wheat continuous futures prices, normalized to 100 at the start of the Bloomberg sample. The series share broad movements but also experience periods of relative divergence.

2 Data Cleaning and Provenance

The raw files were kept unchanged in timestamped directories, while cleaned outputs were written to a separate data layer. Every source was documented with its pull date, file location, and known point-in-time limitations.

For the Bloomberg panel, a date was retained only when both Corn and Wheat prices were present, finite, and positive. We did not forward-fill or interpolate missing prices. We also did not winsorize the price series because the tails are directly relevant to spread and cointegration behaviour. Instead, daily returns above 10 percent in absolute value were flagged and retained for review. In total, 144 Bloomberg rows were removed because at least one leg was missing, non-finite, or non-positive. The exact dates and reasons are recorded in the Bloomberg lineage file.

The main unresolved data issue is the construction of the continuous futures series. Bloomberg supplied a vendor-built continuous contract, but the exact historical roll dates and adjustment rule were not independently reconstructed from individual contracts. This limitation matters because an artificial jump around a roll date can appear as a return or spread signal. We therefore treat the Bloomberg series as the best available research input, not as a perfect reconstruction of tradeable contract history.

3 Features and Models

The target is the next-day return of the Corn to Wheat price ratio:

$$y_{t+1} = \frac{\text{ratio}_{t+1}}{\text{ratio}_t} - 1, \qquad \text{ratio}_t = \frac{\text{Corn}_t}{\text{Wheat}_t}.$$

All features use information available by the close of date t. Rolling windows are trailing and never centred, and the forward target is excluded from the model inputs.

Table 1: Final feature set

Feature	Risk bucket	Purpose
50-day ratio z-score	Statistical	Paper-derived mean-reversion signal
Corn to Wheat ratio	Fundamental analog	Relative price level of the pair
Rolling log spread	Statistical	Cointegration residual using a trailing 60-day hedge ratio
60-day spread z-score	Statistical	Normalized deviation of the rolling spread
20-day Corn volatility	Statistical	Risk-regime measure for the Corn leg
20-day Wheat volatility	Statistical	Risk-regime measure for the Wheat leg
One-day ratio return	Statistical	Short-horizon momentum in the relative price

The paper-derived feature follows van Unen's agricultural-futures pairs-trading methodology. We calculate the Corn to Wheat price ratio, its trailing 50-day mean and standard deviation, and then its rolling z-score:

$$z_t = \frac{\text{ratio}_t - \overline{\text{ratio}}_{t,50}}{s(\text{ratio})_{t,50}}.$$

The paper uses the rolling ratio distribution to identify relative-price divergences. We kept the z-score as a continuous model feature rather than copying the paper's full entry and exit rule.

We estimated two model families. Elastic Net provides a regularized linear benchmark whose coefficients can be interpreted directly. LightGBM provides a nonlinear comparison that can capture interactions between spread, volatility, and momentum features. Validation used five expanding time-series folds on the development sample, with no random shuffling. Elastic Net used an inner three-fold time-series search, while LightGBM used fixed parameters. The target is a one-day forward return computed daily, so adjacent labels overlap and share most of their underlying price information. We kept the data in strict chronological order but chose not to purge overlapping labels across fold boundaries, since doing so would throw out most of the daily observations available in a pairs-trading sample. We are disclosing this label overlap as a known limitation, not presenting it as something we missed.

The cross-validation results show a weak signal rather than a dramatic one. Elastic Net was more consistent and achieved a higher average information coefficient. LightGBM did not produce enough improvement to justify its higher turnover in the portfolio stage.

Table 2: Development-sample time-series cross-validation

Model	Directional accuracy	Spearman IC
Elastic Net	51.45%	0.059
LightGBM	50.68%	0.019

4 Portfolio Construction and Results

A positive prediction creates a long Corn and short Wheat position, while a negative prediction reverses the pair. The Wheat weight is scaled using a trailing 60-day OLS hedge ratio, and the absolute weights are normalized to sum to one. Portfolio weights are lagged by one day so that a signal formed at date t earns returns beginning at date $t+1$.

We evaluated sign, threshold, and capped signal rules, along with an equal-weight model ensemble and two benchmarks, for thirteen pre-registered configurations in total. Transaction costs were modelled at 0, 2, and 5 basis points per unit of turnover. The no-trade threshold was chosen from development-sample out-of-sample predictions before the final holdout was evaluated. Table 3 reports the four most informative strategies; the preferred specification was selected using a stability rule based on sub-period behaviour, drawdown, turnover, and cost robustness together, rather than the single highest holdout Sharpe ratio.

Table 3: Selected holdout results, net of 2 basis points

Strategy	Sharpe	Max DD	Annual turnover	Cum. return
Elastic Net threshold	0.42	-7.5%	7.5	16.1%
Elastic Net sign	-0.12	-31.4%	50.4	-18.2%
LightGBM sign	0.27	-34.8%	82.2	18.4%
Paper z-score benchmark	0.12	-28.0%	22.9	2.6%

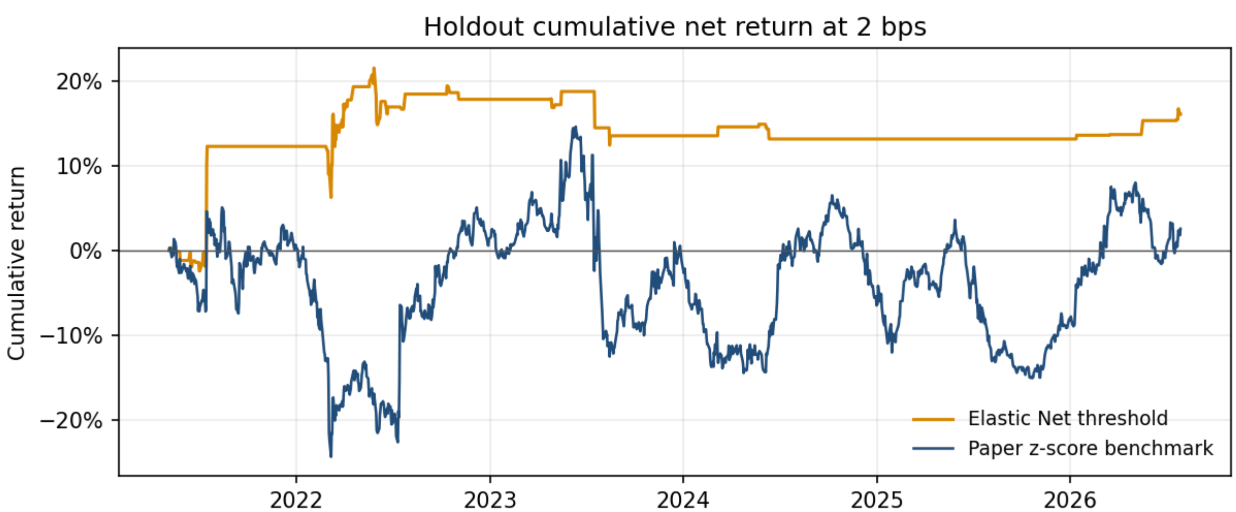

Figure 2: Cumulative holdout return after 2 basis points per unit of turnover. The thresholded Elastic Net strategy trades less frequently and finishes ahead of the paper z-score benchmark over this sample.

The thresholded Elastic Net strategy had the most attractive reported balance of return, drawdown, turnover, and cost sensitivity. Its Sharpe ratio declined only modestly from 0.44 before costs to 0.42 at 2 basis points and 0.39 at 5 basis points. This was mainly due to the no-trade threshold, which prevented the model from acting on small predictions.

The headline result should still be interpreted cautiously. The active directional accuracy was about 51.2 percent, only slightly above a coin flip. Results were also unstable across years, with negative Sharpe ratios in 2023 and

2024. The final holdout was not used to tune features, hyperparameters, or the threshold, but the planned strategies were compared on that holdout to identify the preferred reported specification. For that reason, 0.42 should not be treated as a completely untouched single-test estimate.

5 Residual Diagnostics and Robustness

Elastic Net holdout residuals had a Durbin-Watson statistic of 1.90, which did not suggest strong first-order autocorrelation. Ljung-Box tests rejected white-noise residuals at lags 5 and 10, while the White test rejected constant variance. The Jarque-Bera test strongly rejected normality, showing that the residual distribution remained heavy-tailed.

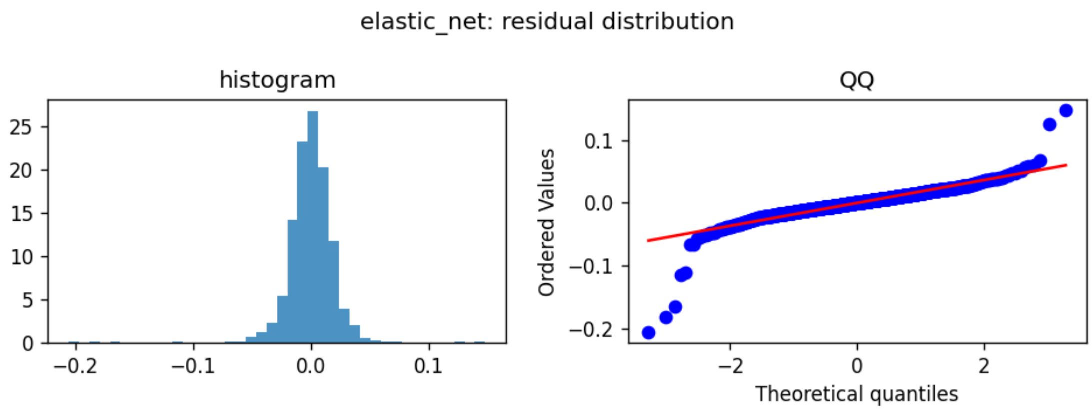

Figure 3: Distribution and normal quantile plot for Elastic Net holdout residuals. The tail departures are consistent with the formal rejection of normality.

The remaining residual structure may reflect volatility clustering, omitted variables, or changing market regimes. We did not add new features after seeing these holdout diagnostics because doing so would reuse the test period. Any heteroskedasticity-aware sizing rule or additional agricultural feature should be treated as a new hypothesis and evaluated on a fresh sample.

Several robustness checks support a cautious interpretation. The preferred strategy remained positive under 40-day, 60-day, and 90-day hedge windows, although the estimated Sharpe changed. Adding one extra day of execution delay reduced the Sharpe to approximately 0.24, suggesting that the signal decays. Most importantly, the deflated Sharpe ratio was approximately zero once we accounted for the roughly thirteen pre-registered configurations we tested (Section 4). The backtest therefore does not establish a statistically reliable discovery.

6 Conclusion

The project found a modest forecasting relationship in the Corn and Wheat relative price, but the practical result depended on trading discipline. Always-on strategies generated too much turnover and performed poorly after costs. A regularized linear model with a no-trade threshold produced the strongest reported holdout profile, with a net Sharpe ratio of about 0.42 and a maximum drawdown of roughly 7.5 percent.

The evidence is promising enough to justify further testing, but not strong enough to claim a durable trading edge. The result varies across years, depends on a vendor-defined continuous futures series, and is not statistically distinguished from the best result expected across multiple trials. The next valid test would use newly observed data and an explicitly reconstructed contract roll. A volatility-aware sizing rule could also be evaluated, provided it is specified before examining the new test period.

Reference

van Unen, Q. (2023). *Pairs Trading in Agricultural Commodity Futures Markets*. BSc thesis, Erasmus University Rotterdam, Section 4.2.